

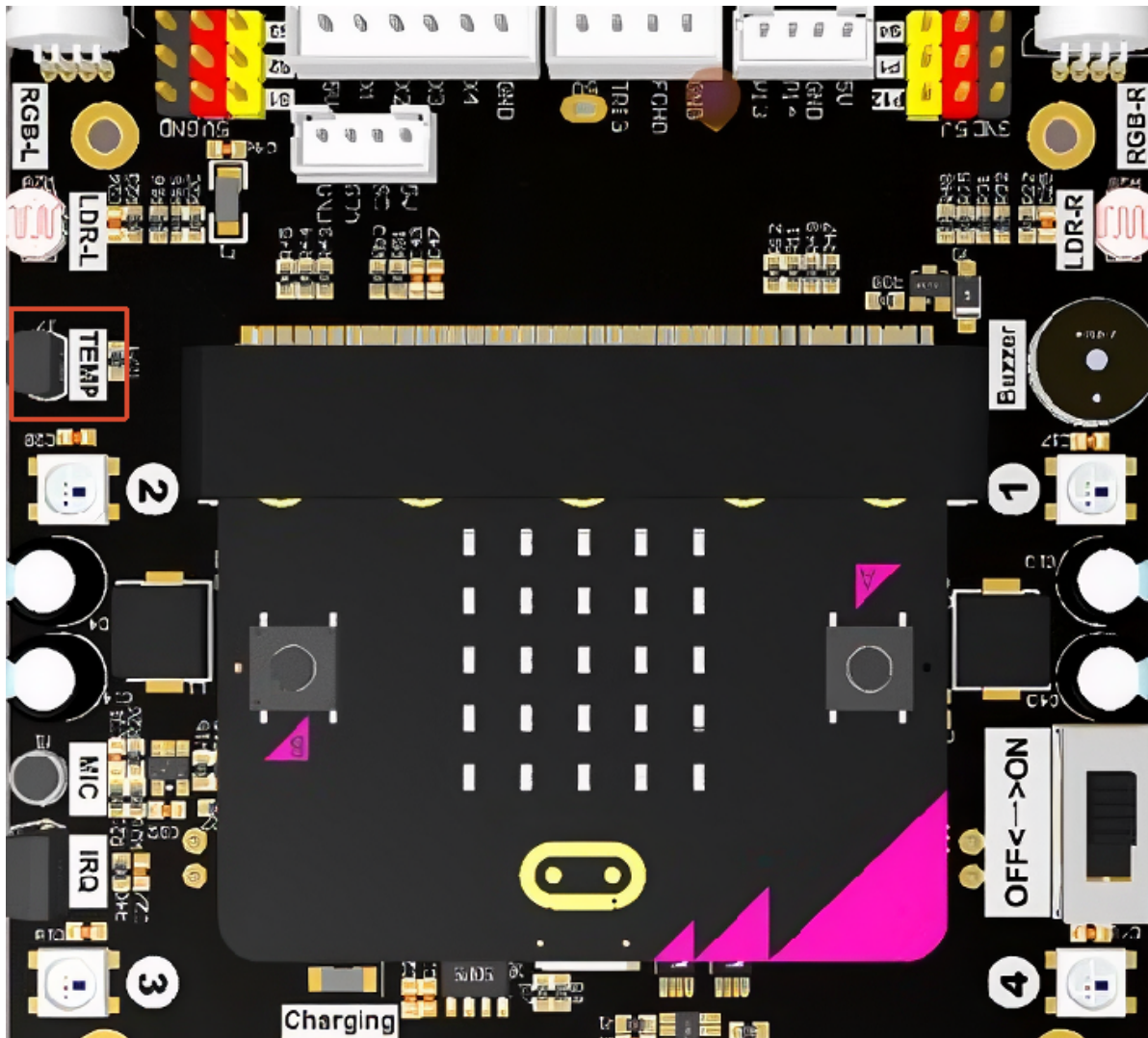
Temperature Collection

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Experimental Purpose

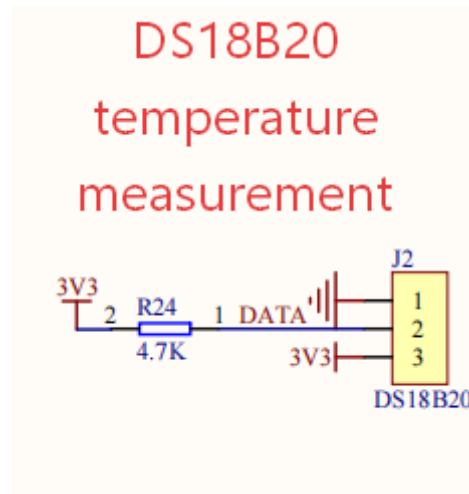
The Lastbit car is equipped with a DS18B20 temperature sensor, marked with the silkscreen number `TEMP`. In this section, we will learn to read the temperature data and display it on the Micro:bit LED matrix.



The DS18B20 is a digital temperature sensor produced by Dallas Semiconductor (Maxim), and its appearance resembles a transistor. Its main features are as follows:

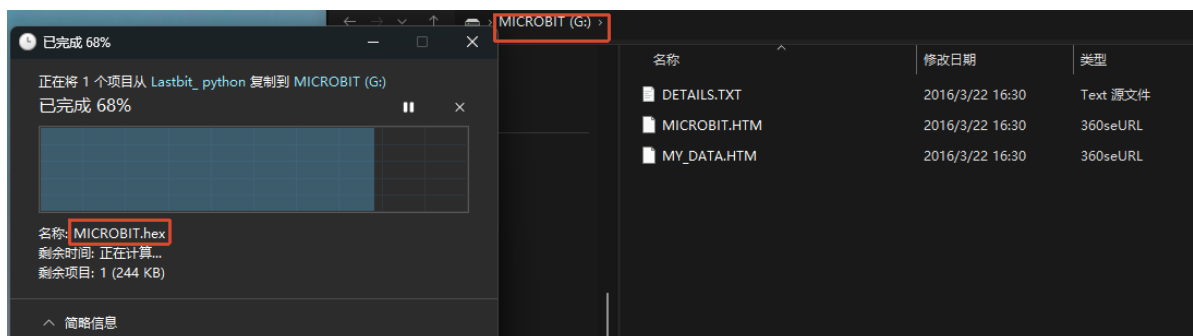
- **Digital signal output:** directly outputs digital temperature values, no ADC conversion required
- **1-Wire single bus communication:** only one data line is needed to communicate with the main controller
- **Measurement range:** -40°C to $+125^{\circ}\text{C}$

- **Measurement accuracy:** $\pm 0.5^{\circ}\text{C}$ (in the range of -10°C to $+85^{\circ}\text{C}$)
- **Programmable resolution:** 9 to 12 bits configurable
- **Unique serial number:** each chip has a unique 64-bit serial number, and multiple sensors can be mounted on the same bus
- **Multi-point networking:** multiple DS18B20s can be connected in parallel on the same data line and read separately
- **Parasitic power supply:** supports drawing power from the data line



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.

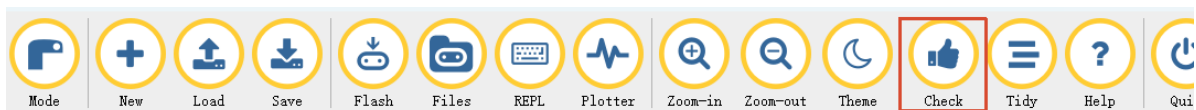
Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Unplug the Microusb data cable and switch the switch to the ON position.

Code Implementation

```
# 5.Temperature_Collection

from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

while True:

    # Read temperature data
    temp = Lastbit.read_temperature()
```

```

if temp is not None:

    # #Show temperature value rounded
    temp = round(temp)
    display.scroll(str(temp))
    # Show the Celsius symbol
    display.show(Image(
        ".##..:"
        "#.###:"
        "#.###:"
        "#.###:"
        "###.."
    ))
    sleep(100)

```

`read_temperature()`: reads the returned temperature value, which is a floating-point number. The unit is degrees Celsius (°C).

`display.show(Image())`: used to display a custom pattern. Here, `.` means the pixel is lit up, and `#` means it is off.

Experimental Phenomenon

Pinch the black probe of the temperature sensor with your hand (**note: do not touch the main board with wet hands to avoid a short circuit**). The Microbit screen will display the read temperature value, which will slowly rise, and after releasing it, the temperature will slowly fall.